```
import pandas as pd
import numpy as np

print("Step 1: Processing CPI Data...")

# Read the Excel file directly
df_cpi_raw = pd.read_excel('CPI full data Base year 2024.xlsx')

# Melt into time-series format
df_cpi = df_cpi_raw.melt(id_vars='Year', var_name='Month', value_name='CPI')

month_map_full = {'January':'01','February':'02','March':'03','April':'04','May':'05','June':'06',
                  'July':'07','August':'08','September':'09','October':'10','November':'11','December':'12'}

df_cpi['Date'] = pd.to_datetime(df_cpi['Year'].astype(str) + '-' + df_cpi['Month'].map(month_map_full) + '-01')
df_cpi_monthly = df_cpi.dropna(subset=['CPI']).set_index('Date').sort_index()[['CPI']]

print("✓ CPI Base 2024 data is ready.")
print(df_cpi_monthly.tail(3))
```

```
Step 1: Processing CPI Data...
✓ CPI Base 2024 data is ready.
               CPI
Date
2025-11-01  104.01
2025-12-01  104.10
2026-01-01  104.46
```

```
print("Step 2: Processing Brent Oil Data...")

df_oil = pd.read_csv('Brent Oil Futures Historical Data Latest.csv')

# Format dates and clean price strings
df_oil['Date'] = pd.to_datetime(df_oil['Date'], format='%d-%m-%Y', errors='coerce')
df_oil['Price'] = pd.to_numeric(df_oil['Price'].astype(str).str.replace(',', ''), errors='coerce')

# Resample to monthly average
df_oil_monthly = df_oil.set_index('Date')[['Price']].resample('MS').mean()
df_oil_monthly.columns = ['Brent_Oil']

print("✓ Brent Oil data is ready.")
print(df_oil_monthly.tail(3))
```

```
Step 2: Processing Brent Oil Data...
✓ Brent Oil data is ready.
            Brent_Oil
Date
2026-02-01  69.735333
2026-03-01  89.602857
```

```
import pandas as pd

print("Step 3: Processing USD/INR Data...")
file_name = 'USD to INR till march 2026.xls'

# Auto-detect the true format of the disguised .xls file
try:
    df_fx = pd.read_excel(file_name, engine='xlrd')
except:
    try:
        df_fx = pd.read_html(file_name)[0]
    except:
        df_fx = pd.read_csv(file_name, sep='\t')
        if len(df_fx.columns) < 2:
            df_fx = pd.read_csv(file_name)

# THE FIX: Force column names to be strings before stripping spaces
df_fx.columns = df_fx.columns.astype(str).str.strip()

# Check if the columns are actually named 'Date' and 'Price'
if 'Date' in df_fx.columns and 'Price' in df_fx.columns:
    df_fx['Date'] = pd.to_datetime(df_fx['Date'], errors='coerce')
    df_fx['Price'] = pd.to_numeric(df_fx['Price'].astype(str).str.replace(',', ''), errors='coerce')

    df_fx_monthly = df_fx.set_index('Date')[['Price']].resample('MS').mean()
    df_fx_monthly.columns = ['USD_INR']

    print("✓ USD/INR data is ready.")
    print(df_fx_monthly.tail(3))
else:
```

```
    print("❌ ERROR: File loaded successfully, but the columns are not named 'Date' and 'Price'.")
    print("Here are the actual column names inside your file:")
    print(df_fx.columns.tolist())
    print("\nPlease look at the list above and update the 'Date' and 'Price' column names in the code.")
```

```
Step 3: Processing USD/INR Data...
/tmp/ipykernel_2316/3062426847.py:11: FutureWarning: Passing literal html to 'read_html' is deprecated and will be removed i
  df_fx = pd.read_html(file_name)[0]
---------------------------------------------------------------------------
FileNotFoundError                         Traceback (most recent call last)
/tmp/ipykernel_2316/3062426847.py in <cell line: 0>()
      7 try:
----> 8     df_fx = pd.read_excel(file_name, engine='xlrd')
      9 except:

                              16 frames
FileNotFoundError: [Errno 2] No such file or directory: 'USD to INR till march 2026.xls'

During handling of the above exception, another exception occurred:

ValueError                                Traceback (most recent call last)
ValueError: No tables found

During handling of the above exception, another exception occurred:

FileNotFoundError                         Traceback (most recent call last)
/usr/local/lib/python3.12/dist-packages/pandas/io/common.py in get_handle(path_or_buf, mode, encoding, compression,
memory_map, is_text, errors, storage_options)
    871         if ioargs.encoding and "b" not in ioargs.mode:
    872             # Encoding
--> 873             handle = open(
    874                 handle,
    875                 ioargs.mode,

FileNotFoundError: [Errno 2] No such file or directory: 'USD to INR till march 2026.xls'
```

```
import pandas as pd
import numpy as np
from google.colab import files

print("🚀 Starting Master Dataset ETL Pipeline...\n")

# --- 1. CPI Data ---
df_cpi_raw = pd.read_excel('CPI full data Base year 2024.xlsx')
df_cpi = df_cpi_raw.melt(id_vars='Year', var_name='Month', value_name='CPI')
month_map_full = {'January':'01','February':'02','March':'03','April':'04','May':'05','June':'06',
                  'July':'07','August':'08','September':'09','October':'10','November':'11','December':'12'}
df_cpi['Date'] = pd.to_datetime(df_cpi['Year'].astype(str) + '-' + df_cpi['Month'].map(month_map_full) + '-01')
df_cpi_monthly = df_cpi.dropna(subset=['CPI']).set_index('Date').sort_index()[['CPI']]
print("✓ CPI Base 2024 data is ready.")

# --- 2. Brent Oil Data ---
df_oil = pd.read_csv('Brent Oil Futures Historical Data Latest.csv')
df_oil['Date'] = pd.to_datetime(df_oil['Date'], format='%d-%m-%Y', errors='coerce')
df_oil['Price'] = pd.to_numeric(df_oil['Price'].astype(str).str.replace(',', ''), errors='coerce')
df_oil_monthly = df_oil.set_index('Date')[['Price']].resample('MS').mean()
df_oil_monthly.columns = ['Brent_Oil']
print("✓ Brent Oil data is ready.")

# --- 3. USD/INR Data (USING THE CLEAN CSV!) ---
df_fx = pd.read_csv('usd to inr march.csv')
df_fx = df_fx.iloc[:, [0, 1]]
df_fx.columns = ['Date', 'Price']
df_fx['Date'] = pd.to_datetime(df_fx['Date'], errors='coerce')
df_fx['Price'] = pd.to_numeric(df_fx['Price'].astype(str).str.replace(',', ''), errors='coerce')
df_fx_monthly = df_fx.dropna(subset=['Date', 'Price']).set_index('Date')[['Price']].resample('MS').mean()
df_fx_monthly.columns = ['USD_INR']
print("✓ USD/INR data is ready.")

# --- 4. GDP Data ---
df_gdp = pd.read_csv('gdp qaurterly only the gdp data.csv')
def q_to_date(row):
    year = int(str(row['Financial_Year'])[:4])
    q_map = {'Q1':'-04-01', 'Q2':'-07-01', 'Q3':'-10-01', 'Q4':'-01-01'}
    yr = year if row['Quarter'] != 'Q4' else year+1
    return pd.to_datetime(str(yr) + q_map[row['Quarter']])
df_gdp['Date'] = df_gdp.apply(q_to_date, axis=1)
df_gdp_monthly = df_gdp.set_index('Date')[['GDP_Market_Prices_Crores']].resample('MS').interpolate(method='linear')
df_gdp_monthly.columns = ['GDP']
print("✓ GDP data is ready.")

# --- 5. M3 Money Supply Data ---
df_m3 = pd.read_csv('M3 Latest.csv')
```

```
month_map_num = {'January': 1, 'February': 2, 'March': 3, 'April': 4, 'May': 5, 'June': 6,
                 'July': 7, 'August': 8, 'September': 9, 'October': 10, 'November': 11, 'December': 12}
def parse_m3(row):
    try:
        year_val = int(str(row['Year'])[:4])
        month_str = str(row['Month']).strip()
        if month_str in ['January', 'February', 'March']: year_val += 1
        return pd.to_datetime(f"{year_val}-{month_map_num[month_str]:02d}-01")
    except: return pd.NaT
df_m3['Date'] = df_m3.apply(parse_m3, axis=1)
df_m3_monthly = df_m3.dropna(subset=['Date']).sort_values('Date').set_index('Date')[['M3_Broad_Money']]
df_m3_monthly.columns = ['M3_Money_Supply']
print("✓ M3 data is ready.")

# --- 6. Master Merge & Export ---
print("\nStep 6: Merging Master Dataset...")
master_df = df_cpi_monthly.join([df_oil_monthly, df_fx_monthly, df_gdp_monthly, df_m3_monthly], how='outer')
master_df = master_df.loc['2013-01-01':]
feature_cols = ['Brent_Oil', 'USD_INR', 'GDP', 'M3_Money_Supply']
master_df[feature_cols] = master_df[feature_cols].interpolate(method='linear').ffill().bfill()

output_file = 'Indian_Inflation_Master_Data_Base2024_Final.csv'
master_df.to_csv(output_file)
files.download(output_file)

print(f"🏁 SUCCESS: '{output_file}' has been created and is downloading.")
print("\n--- FINAL DATASET PREVIEW ---")
print(master_df.tail(5))
```

```
🚀 Starting Master Dataset ETL Pipeline...

✓ CPI Base 2024 data is ready.
✓ Brent Oil data is ready.
✓ USD/INR data is ready.
✓ GDP data is ready.
✓ M3 data is ready.

Step 6: Merging Master Dataset...
🏁 SUCCESS: 'Indian_Inflation_Master_Data_Base2024_Final.csv' has been created and is downloading.

--- FINAL DATASET PREVIEW ---
            CPI  Brent_Oil    USD_INR        GDP  M3_Money_Supply
Date
2026-08-01  NaN      86.66  89.893000  8525069.0       29903539.0
2026-09-01  NaN      86.66  90.934667  8525069.0       29903539.0
2026-10-01  NaN      86.66  90.577300  8525069.0       29903539.0
2026-11-01  NaN      86.66  90.567500  8525069.0       29903539.0
2026-12-01  NaN      86.66  90.392300  8525069.0       29903539.0
```

Start coding or generate with AI.